

2026 年 06 月 CSCA 官方数学试卷

CSCA Official Mathematics Examination Paper, June 2026

第 1 题 Question 1

若集合 $A = \{x|x^2 = 1\}$, 则下列表示正确的是 ()。

If the set $A = \{x|x^2 = 1\}$, which of the following expressions is correct?

A. $A = -1$

B. $\{1\} \subseteq A$

C. $1 \subseteq A$

D. $A = \{1\}$

答案 Answer: B

解析 Analysis:

中文: 解方程 $x^2 = 1$ 得 $x = \pm 1$, 因此集合 $A = \{-1, 1\}$ 。 $\{1\}$ 是 A 的子集, 符号 $\subseteq$ 表示集合间的包含关系, 表述正确; 元素与集合的关系应使用 $\in$ 符号。

英文: Solve $x^2 = 1$ and we get $x = \pm 1$, so $A = \{-1, 1\}$. $\{1\}$ is a subset of A , and the symbol $\subseteq$ correctly represents the inclusion relation between sets; the symbol $\in$ should be used for the relation between elements and sets.

第 2 题 Question 2

若集合 $A = \{1, 2, 3\}$, $B = \{2, 3, \dots, 2026\}$, 则 $A \cap B = ()$ 。

If sets $A = \{1, 2, 3\}$ and $B = \{2, 3, \dots, 2026\}$, then $A \cap B = ()$.

A. $\{2, 3\}$

B. $\{1, 2, 3\}$

C. $\{2, 3, \dots, 2026\}$

D. $\{1, 2, \dots, 2026\}$

答案 Answer: A

解析 Analysis:

中文: 集合的交集是由两个集合的所有公共元素组成的集合, A 和 B 的公共元素为 2 和 3, 因此 $A \cap B = \{2, 3\}$ 。

英文: The intersection of two sets consists of all their common elements. The common elements of A and B are 2 and 3, so $A \cap B = \{2, 3\}$.

第 3 题 Question 3

一元二次不等式 $3(x+4)(x-2) < 0$ 的解集为 ()。

The solution set of the quadratic inequality in one variable $3(x+4)(x-2) < 0$ is () .

- A. $(2, +\infty)$
- B. $(-4, 2)$
- C. $(-\infty, -4) \cup (2, +\infty)$
- D. $(-\infty, 2) \cup (-4, +\infty)$

答案 Answer: B

解析 Analysis:

中文：二次函数 $y = 3(x+4)(x-2)$ 开口向上，零点为 $x = -4$ 和 $x = 2$ ，函数值小于 0 的区间在两根之间，因此解集为 $(-4, 2)$ 。

英文：The quadratic function $y = 3(x+4)(x-2)$ opens upward with zeros at $x = -4$ and $x = 2$. The interval where the function value is less than 0 lies between the two roots, so the solution set is $(-4, 2)$.

第 4 题 Question 4

函数 $y = \frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x-3}$ 的定义域为 ()。

The domain of the function $y = \frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x-3}$ is () .

- A. $(-\infty, 1) \cup (1, 2) \cup (2, 3) \cup (3, +\infty)$
- B. $\{1, 2, 3\}$
- C. $[1, 3]$
- D. $\mathbb{R}$

答案 Answer: A

解析 Analysis:

中文：分式的分母不能为 0，因此 $x - 1 \neq 0$ ， $x - 2 \neq 0$ ， $x - 3 \neq 0$ ，即 $x \neq 1, 2, 3$ ，定义域为全体实数去掉这三个点。

英文：The denominator of a fraction cannot be zero, so $x \neq 1$, $x \neq 2$ and $x \neq 3$. The domain is all real numbers except these three points.

第 5 题 Question 5

若点 $P(x, y)$ 位于第二象限，则点 $Q(2x, 3y)$ 位于 ()。

If point $P(x, y)$ lies in Quadrant II, then point $Q(2x, 3y)$ lies in () .

- A. 第四象限 (Quadrant IV)
- B. 第三象限 (Quadrant III)
- C. 第二象限 (Quadrant II)
- D. 第一象限 (Quadrant I)

答案 Answer: C

$f(x)$, and its graph is symmetric about the origin. $f(-x) = f(x)$ is the property of an even function, not an odd function.

第 8 题 Question 8

函数 $y = 2x$, $x \in [-1, 1]$ 的反函数是 ()。

The inverse function of $y = 2x$, $x \in [-1, 1]$ is () .

A. $y = \frac{x}{2}, x \in [-1, 1]$

B. $y = \frac{x}{2}$

C. $y = \frac{x}{2}, x \in [-2, 2]$

D. $y = 2x, x \in [-1, 1]$

答案 Answer: C

解析 Analysis:

中文：原函数的值域为 $[-2, 2]$ ，即反函数的定义域。由 $y = 2x$ 反解得 $x = \frac{y}{2}$ ，交换 x, y 得反函数 $y = \frac{x}{2}$ ，定义域为 $[-2, 2]$ 。

英文：The range of the original function is $[-2, 2]$, which is the domain of the inverse function. Solve $y = 2x$ for x to get $x = \frac{y}{2}$, swap x and y , and the inverse function is $y = \frac{x}{2}$ with domain $[-2, 2]$.

第 9 题 Question 9

下列函数中单调递减的是 ()。

Which of the following functions is monotonically decreasing?

A. $y = e^{-x}$

B. $y = x^2 - 1$

C. $y = 100^x$

D. $y = \cos x$

答案 Answer: A

解析 Analysis:

中文： $y = e^{-x} = (\frac{1}{e})^x$ ，底数 $0 < \frac{1}{e} < 1$ ，是定义域上单调递减的指数函数；其余选项均不是定义域上的单调递减函数。

英文： $y = e^{-x} = (\frac{1}{e})^x$ is an exponential function with base $0 < \frac{1}{e} < 1$, so it is monotonically decreasing over its domain. The other options are not monotonically decreasing over their domains.

第 10 题 Question 10

分式不等式 $\frac{-2}{x-2} \leq 0$ 的解集为 ()。

The solution set of the fractional inequality $\frac{-2}{x-2} \leq 0$ is ().

- A. $(-\infty, 2)$
- B. $(2, +\infty)$
- C. $[0, 2]$
- D. $(-\infty, 2]$

答案 Answer: B

解析 Analysis:

中文：分子 $-2 < 0$ ，要使分式小于等于 0，需分母 $x - 2 > 0$ ；分母不能为 0，且分子恒不为 0，因此无法取到等号，解得 $x > 2$ ，解集为 $(2, +\infty)$ 。

英文：The numerator $-2 < 0$. For the fraction to be less than or equal to 0, the denominator must satisfy $x - 2 > 0$. The denominator cannot be zero, and the numerator is never zero, so the equality cannot hold. Thus $x > 2$, and the solution set is $(2, +\infty)$.

第 11 题 Question 11

下列点位于第三象限的是 ()。

Which of the following points lies in Quadrant III?

- A. $(-1, -2)$
- B. $(-1, 2)$
- C. $(1, 2)$
- D. $(1, -2)$

答案 Answer: A

解析 Analysis:

中文：第三象限内的点横坐标和纵坐标均为负数，点 $(-1, -2)$ 满足横、纵坐标均负，位于第三象限。

英文：Points in Quadrant III have both negative x-coordinate and negative y-coordinate. Point $(-1, -2)$ satisfies this condition and lies in Quadrant III.

第 12 题 Question 12

已知 a, b 和 c 构成等比数列并且满足 $ac = 100$ ，则 $b = ()$ 。

Given that a, b and c form a geometric sequence and satisfy $ac = 100$, then $b = ()$.

- A. ± 100
- B. 10
- C. 100
- D. ± 10

答案 Answer: D

解析 Analysis:

中文：由等比中项性质， $b^2 = ac = 100$ ；等比中项可取正负两值，开方得 $b = \pm 10$ 。

英文：By the property of geometric mean, $b^2 = ac = 100$. The geometric mean can take both positive and negative values. Taking the square root gives $b = \pm 10$.

第 13 题 Question 13

若直线经过点 $(-1,1)$ 和 $(0,2)$ ，则直线的倾斜角为 ()。

If a straight line passes through points $(-1,1)$ and $(0,2)$, the angle of inclination of the line is ().

A. 135°

B. 30°

C. 45°

D. 60°

答案 Answer: C

解析 Analysis:

中文：由斜率公式得 $k = \frac{2-1}{0-(-1)} = 1$ ，设倾斜角为 θ ，则 $\tan \theta = 1$ ，且

$0^\circ \leq \theta < 180^\circ$ ，因此 $\theta = 45^\circ$ 。

英文：By the slope formula, $k = \frac{2-1}{0-(-1)} = 1$. Let the inclination angle be θ , then $\tan \theta = 1$ and $0^\circ \leq \theta < 180^\circ$, so $\theta = 45^\circ$.

第 14 题 Question 14

若点 $P(1,2)$ 是角 α 的终边上的点，则 $\sin \alpha = ()$ 。

If point $P(1,2)$ is on the terminal side of angle α , then $\sin \alpha = ()$.

A. $\frac{\sqrt{5}}{5}$

B. $\frac{\sqrt{2}}{2}$

C. $\frac{1}{2}$

D. $\frac{2\sqrt{5}}{5}$

答案 Answer: D

解析 Analysis:

中文：点到原点的距离 $r = \sqrt{1^2 + 2^2} = \sqrt{5}$ ，由三角函数定义 $\sin \alpha = \frac{y}{r} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$ 。

英文：The distance from the point to the origin is $r = \sqrt{1^2 + 2^2} = \sqrt{5}$. By definition of trigonometric functions, $\sin \alpha = \frac{y}{r} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$.

第 15 题 Question 15

经过点 $A(-1,1)$, $B(2,3)$ 的直线的斜率为 ()。

The slope of the straight line passing through points $A(-1,1)$ and $B(2,3)$ is ().

- A. $\frac{2}{3}$
- B. $-\frac{2}{3}$
- C. $\frac{3}{2}$
- D. $-\frac{3}{2}$

答案 Answer: A

解析 Analysis:

中文：由两点间斜率公式， $k = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{2 - (-1)} = \frac{2}{3}$ 。

英文：By the two-point slope formula, $k = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{2 - (-1)} = \frac{2}{3}$.

第 16 题 Question 16

已知 $\{a_n\}$ 是等差数列，且 $a_1 = 1$, $d = 2$ ，则 $a_9 = ()$ 。

Given $\{a_n\}$ is an arithmetic sequence with $a_1 = 1$ and $d = 2$, then $a_9 = ()$.

- A. 20
- B. 18
- C. 19
- D. 17

答案 Answer: D

解析 Analysis:

中文：等差数列通项公式为 $a_n = a_1 + (n - 1)d$ ，代入得 $a_9 = 1 + 8 \times 2 = 17$ 。

英文：The general term of an arithmetic sequence is $a_n = a_1 + (n - 1)d$. Substitute the values: $a_9 = 1 + 8 \times 2 = 17$.

第 17 题 Question 17

点 $P(a, b)$ 到点 $Q(a + 1, b - 1)$ 的距离为 ()。

The distance between point $P(a, b)$ and point $Q(a + 1, b - 1)$ is ().

- A. $\sqrt{(2a + 1)^2 + (2b - 1)^2}$
- B. 2
- C. $\sqrt{2}$
- D. 4

答案 Answer: C

解析 Analysis:

中文：由两点间距离公式，距离

$$d = \sqrt{(a + 1 - a)^2 + (b - 1 - b)^2} = \sqrt{1^2 + (-1)^2} = \sqrt{2}。$$

英文：By the distance formula between two points, $d = \sqrt{(a+1-a)^2 + (b-1-b)^2} = \sqrt{1^2 + (-1)^2} = \sqrt{2}$.

第 18 题 Question 18

过原点并且斜率为 2 的直线方程是 ()。

The equation of the straight line passing through the origin with a slope of 2 is ().

- A. $y = x + 2$
- B. $y = x$
- C. $y = 2x$
- D. $y = 2x + 2$

答案 Answer: C

解析 Analysis:

中文：由直线的斜截式方程 $y = kx + b$ ，过原点则截距 $b = 0$ ，斜率 $k = 2$ ，因此方程为 $y = 2x$ 。

英文：Using the slope-intercept form $y = kx + b$, since the line passes through the origin, the intercept $b = 0$. With slope $k = 2$, the equation is $y = 2x$.

第 19 题 Question 19

圆心为 (2,5)，半径为 5 的圆的方程是 ()。

The equation of a circle with center (2,5) and radius 5 is ().

- A. $(x - 2)^2 + (y - 5)^2 = 5$
- B. $(x - 2)^2 + (y - 5)^2 = 25$
- C. $(x + 2)^2 + (y + 5)^2 = 25$
- D. $(x - 5)^2 + (y - 2)^2 = 25$

答案 Answer: B

解析 Analysis:

中文：圆的标准方程为 $(x - h)^2 + (y - k)^2 = r^2$ ，其中 (h, k) 为圆心， r 为半径，代入得 $(x - 2)^2 + (y - 5)^2 = 25$ 。

英文：The standard equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius. Substitution gives $(x - 2)^2 + (y - 5)^2 = 25$.

第 20 题 Question 20

若 $\cos^2 \alpha = \frac{1}{2}$ ，则 $\tan \alpha = ()$ 。

If $\cos^2 \alpha = \frac{1}{2}$, then $\tan \alpha = ()$.

- A. 1
- B. ± 1

- C. 2
D. $\frac{1}{2}$

答案 Answer: B

解析 Analysis:

中文：由 $\cos^2 \alpha = \frac{1}{2}$ 得 $\cos \alpha = \pm \frac{\sqrt{2}}{2}$ ，对应 $\sin \alpha = \pm \frac{\sqrt{2}}{2}$ ，因此 $\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \pm 1$ 。

英文：From $\cos^2 \alpha = \frac{1}{2}$, we get $\cos \alpha = \pm \frac{\sqrt{2}}{2}$ and correspondingly $\sin \alpha = \pm \frac{\sqrt{2}}{2}$. Thus $\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \pm 1$.

第 21 题 Question 21

已知等比数列 $\{a_n\}$ 的通项公式为 $a_n = 2 \times (-1)^{n+1}$ ，则 ()。

Given the general term formula of the geometric sequence $\{a_n\}$ is $a_n = 2 \times (-1)^{n+1}$, then ().

- A. 数列的公比 $q = -2$ (The common ratio of the sequence $q = -2$)
B. 数列的首项 $a_1 = -2$ (The first term of the sequence $a_1 = -2$)
C. 数列的第 100 项 $a_{100} = 2$ (The 100th term of the sequence $a_{100} = 2$)
D. 数列的前 10 项和等于 0 (The sum of the first 10 terms of the sequence is equal to 0)

答案 Answer: D

解析 Analysis:

中文：该等比数列首项 $a_1 = 2$ ，公比 $q = -1$ ； $a_{100} = 2 \times (-1)^{101} = -2$ ；公比为 -1 时，前偶数项的和为 0，因此前 10 项和为 0。

英文：The first term is $a_1 = 2$ and the common ratio is $q = -1$. $a_{100} = 2 \times (-1)^{101} = -2$. For a geometric sequence with $q = -1$, the sum of the first even number of terms is 0, so the sum of the first 10 terms is 0.

第 22 题 Question 22

已知 $a > b$, $c \in \mathbb{R}$ ，则 ()。

Given $a > b$ and $c \in \mathbb{R}$, then ().

- A. $a^2 > b^2$
B. $ac > bc$
C. $a + c > b + c$
D. $|a| > |b|$

答案 Answer: C

解析 Analysis:

中文：根据不等式的基本性质，不等式两边同时加上同一个实数，不等号方向不变，

因此 $a + c > b + c$ 恒成立。

英文: By the basic property of inequalities, adding the same real number to both sides does not change the direction of the inequality sign, so $a + c > b + c$ always holds.

第 23 题 Question 23

下列关于函数 $y = x^2$ 说法正确的是 ()。

Which of the following statements about the function $y = x^2$ is correct?

- A. 该函数的图像关于原点对称 (The graph of the function is symmetric about the origin)
- B. 该函数在定义域上单调递减 (The function is monotonically decreasing over its domain)
- C. 该函数的图像关于 y 轴对称 (The graph of the function is symmetric about the y -axis)
- D. 该函数在定义域上单调递增 (The function is monotonically increasing over its domain)

答案 Answer: C

解析 Analysis:

中文: $y = x^2$ 是偶函数, 图像关于 y 轴对称; 它在 $(-\infty, 0)$ 单调递减, 在 $(0, +\infty)$ 单调递增, 在整个定义域上不单调。

英文: $y = x^2$ is an even function whose graph is symmetric about the y -axis. It decreases on $(-\infty, 0)$ and increases on $(0, +\infty)$, and is not monotonic over the whole domain.

第 24 题 Question 24

若数列 $\{a_n\}$ 是等差数列, 且 $a_1 = 1$, $a_2 = 3$, 则 $a_5 = ()$ 。

If the sequence $\{a_n\}$ is an arithmetic sequence with $a_1 = 1$ and $a_2 = 3$, then $a_5 = ()$.

- A. 11
- B. 5
- C. 9
- D. 7

答案 Answer: C

解析 Analysis:

中文: 公差 $d = a_2 - a_1 = 2$, 由通项公式得 $a_5 = a_1 + 4d = 1 + 4 \times 2 = 9$ 。

英文: The common difference is $d = a_2 - a_1 = 2$. By the general term formula, $a_5 = a_1 + 4d = 1 + 4 \times 2 = 9$.

第 25 题 Question 25

若 $\sin^2 \alpha = \frac{1}{4}$, 则 $\cos 2\alpha = ()$ 。

If $\sin^2 \alpha = \frac{1}{4}$, then $\cos 2\alpha = ()$.

A. $-\frac{7}{8}$

B. $\frac{7}{8}$

C. $\frac{1}{2}$

D. $-\frac{1}{2}$

答案 Answer: C

解析 Analysis:

中文: 由二倍角公式 $\cos 2\alpha = 1 - 2\sin^2 \alpha$, 代入得 $\cos 2\alpha = 1 - 2 \times \frac{1}{4} = \frac{1}{2}$ 。

英文: By the double-angle formula $\cos 2\alpha = 1 - 2\sin^2 \alpha$, substitute to get

$$\cos 2\alpha = 1 - 2 \times \frac{1}{4} = \frac{1}{2}.$$

第 26 题 Question 26

已知直线 $l_1: y = x + 3$, 直线 l_2 经过 $A(1,0)$ 和 $B(0,1)$, 则直线 l_1 和直线 l_2 的交点为 $()$ 。

Given line $l_1: y = x + 3$, and line l_2 passes through $A(1,0)$ and $B(0,1)$, the intersection point of line l_1 and line l_2 is $()$.

A. $(-\frac{3}{2}, \frac{3}{2})$

B. $(-2,1)$

C. $(-1,2)$

D. $(0,3)$

答案 Answer: C

解析 Analysis:

中文: 先求得 l_2 的方程为 $y = -x + 1$, 联立方程组 $\begin{cases} y = x + 3 \\ y = -x + 1 \end{cases}$, 解得 $x = -1$, $y = 2$, 交点为 $(-1,2)$ 。

英文: The equation of l_2 is $y = -x + 1$. Solve the system $\begin{cases} y = x + 3 \\ y = -x + 1 \end{cases}$ to get $x = -1$, $y = 2$. The intersection point is $(-1,2)$.

第 27 题 Question 27

若圆过点 $A(2,2)$, 圆心为 $(2,0)$, 则该圆的方程为 $()$ 。

If a circle passes through point $A(2,2)$ with center $(2,0)$, the equation of the circle is $()$.

A. $(x-2)^2 + (y-2)^2 = 4$

B. $(x-2)^2 + y^2 = 2$

C. $(x-2)^2 + y^2 = 4$

D. $x^2 + (y - 2)^2 = 2$

答案 Answer: C

解析 Analysis:

中文：半径为圆心到点 A 的距离， $r = \sqrt{(2-2)^2 + (2-0)^2} = 2$ ，代入圆的标准方程得 $(x-2)^2 + y^2 = 4$ 。

英文：The radius is the distance from the center to point A : $r = \sqrt{(2-2)^2 + (2-0)^2} = 2$. Substitute into the standard equation: $(x-2)^2 + y^2 = 4$.

第 28 题 Question 28

双曲线 $\frac{x^2}{16} - \frac{y^2}{9} = 1$ 的焦点为 ()。

The foci of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ are () .

A. $(\pm 25, 0)$

B. $(0, \pm 25)$

C. $(0, \pm 5)$

D. $(\pm 5, 0)$

答案 Answer: D

解析 Analysis:

中文：双曲线焦点在 x 轴上， $a^2 = 16$ ， $b^2 = 9$ ，由 $c^2 = a^2 + b^2 = 25$ 得 $c = 5$ ，因此焦点坐标为 $(\pm 5, 0)$ 。

英文：The foci are on the x -axis. With $a^2 = 16$, $b^2 = 9$, we have $c^2 = a^2 + b^2 = 25$, so $c = 5$. The foci are $(\pm 5, 0)$.

第 29 题 Question 29

计算 $\frac{\log_5 8}{\log_{25} 4} + \log_5 1 = ()$ 。

Evaluate $\frac{\log_5 8}{\log_{25} 4} + \log_5 1 = ()$.

A. 3

B. 2

C. 0

D. 4

答案 Answer: A

解析 Analysis:

中文：由换底公式， $\log_{25} 4 = \log_5 2$ ， $\log_5 8 = 3\log_5 2$ ，因此分式值为 3；又 $\log_5 1 = 0$ ，原式总和为 3。

英文：By the change-of-base formula, $\log_{25} 4 = \log_5 2$ and $\log_5 8 = 3\log_5 2$, so the fraction equals 3. Since $\log_5 1 = 0$, the total value of the expression is 3.

第 30 题 Question 30

若数列 $\{a_n\}$ 是 $\frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, \dots$, 则通项 $a_n = ()$ 。

If the sequence $\{a_n\}$ is $\frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, \dots$, then the general term $a_n = ()$.

- A. $\frac{1}{2^{n-1}}$
- B. $\frac{1}{2^n}$
- C. $(-1)^{n-1} \frac{1}{2^n}$
- D. $(-1)^n \frac{1}{2^n}$

答案 Answer: C

解析 Analysis:

中文: 该数列是首项为 $\frac{1}{2}$ 、公比为 $-\frac{1}{2}$ 的等比数列, 通项为 $a_n = \frac{1}{2} \cdot (-\frac{1}{2})^{n-1} = (-1)^{n-1} \frac{1}{2^n}$ 。

英文: This is a geometric sequence with first term $\frac{1}{2}$ and common ratio $-\frac{1}{2}$. The general term is $a_n = \frac{1}{2} \cdot (-\frac{1}{2})^{n-1} = (-1)^{n-1} \frac{1}{2^n}$.

第 31 题 Question 31

下列关于函数 $y = \cos x$ 说法正确的是 ()。

Which of the following statements about the function $y = \cos x$ is correct?

- A. 该函数的图像关于原点对称 (The graph of the function is symmetric about the origin)
- B. 该函数有无穷多的零点 (The function has infinitely many zeros)
- C. $\cos \pi = \cos 2\pi$
- D. 该函数在 $[2\pi, 3\pi]$ 上单调上升 (The function is monotonically increasing on $[2\pi, 3\pi]$)

答案 Answer: B

解析 Analysis:

中文: $\cos x = 0$ 的解为 $x = \frac{\pi}{2} + k\pi (k \in \mathbb{Z})$, 有无穷多个, 因此函数有无穷多零点; 其余选项均错误。

英文: The solutions of $\cos x = 0$ are $x = \frac{\pi}{2} + k\pi (k \in \mathbb{Z})$, which are infinitely many, so the function has infinitely many zeros. The other options are incorrect.

第 32 题 Question 32

关于抛物线 $y = \frac{x^2}{4}$, 下列说法正确的是 ()。

Which of the following statements about the parabola $y = \frac{x^2}{4}$ is correct?

- A. 对称轴为 $y = 0$ (The axis of symmetry is $y = 0$)
 B. 该抛物线通过点 $(4, -4)$ (The parabola passes through point $(4, -4)$)
 C. 焦点坐标为 $(0, \pm 1)$ (The focus coordinates are $(0, \pm 1)$)
 D. 准线方程为 $y = -1$ (The directrix equation is $y = -1$)

答案 Answer: D

解析 Analysis:

中文：抛物线标准形式为 $x^2 = 4y$ ，开口向上，对称轴为 $x = 0$ ，焦点为 $(0, 1)$ ，准线方程为 $y = -1$ 。

英文：The standard form is $x^2 = 4y$, opening upward. Its axis of symmetry is $x = 0$, focus is $(0, 1)$, and directrix is $y = -1$.

第 33 题 Question 33

关于三条直线 $l_1: x + 2y + 1 = 0$, $l_2: x - y = 0$, $l_3: 2x - y + 8 = 0$ ，下列说法正确的是 ()。

Regarding the three straight lines $l_1: x + 2y + 1 = 0$, $l_2: x - y = 0$, $l_3: 2x - y + 8 = 0$, which of the following statements is correct?

- A. l_1 与 l_3 平行 (l_1 is parallel to l_3)
 B. l_1 与 l_2 垂直 (l_1 is perpendicular to l_2)
 C. l_1 与 l_2 平行 (l_1 is parallel to l_2)
 D. l_1 与 l_3 垂直 (l_1 is perpendicular to l_3)

答案 Answer: D

解析 Analysis:

中文：三条直线斜率分别为 $k_1 = -\frac{1}{2}$, $k_2 = 1$, $k_3 = 2$; $k_1 \cdot k_3 = -1$ ，因此 l_1 与 l_3 垂直。

英文：The slopes are $k_1 = -\frac{1}{2}$, $k_2 = 1$, $k_3 = 2$. Since $k_1 \cdot k_3 = -1$, l_1 is perpendicular to l_3 .

第 34 题 Question 34

已知 $\tan \frac{\pi}{4} = 1$, $\tan \frac{\pi}{6} = \frac{\sqrt{3}}{3}$, 则 $\tan \frac{\pi}{12} = ()$ 。

Given $\tan \frac{\pi}{4} = 1$ and $\tan \frac{\pi}{6} = \frac{\sqrt{3}}{3}$, then $\tan \frac{\pi}{12} = ()$.

- A. $1 - \frac{\sqrt{3}}{3}$
 B. $1 + \frac{\sqrt{3}}{3}$
 C. $2 - \sqrt{3}$
 D. $2 + \sqrt{3}$

答案 Answer: C

解析 Analysis:

中文：利用正切差角公式， $\tan \frac{\pi}{12} = \tan \left(\frac{\pi}{4} - \frac{\pi}{6} \right) = \frac{\tan \frac{\pi}{4} - \tan \frac{\pi}{6}}{1 + \tan \frac{\pi}{4} \tan \frac{\pi}{6}}$ ，代入化简得 $2 - \sqrt{3}$ 。

英文：Using the tangent subtraction formula, $\tan \frac{\pi}{12} = \tan \left(\frac{\pi}{4} - \frac{\pi}{6} \right) = \frac{\tan \frac{\pi}{4} - \tan \frac{\pi}{6}}{1 + \tan \frac{\pi}{4} \tan \frac{\pi}{6}}$. Simplification gives $2 - \sqrt{3}$.

第 35 题 Question 35

函数 $y = \frac{\sqrt{x+3}}{x-2} + 3\log x$ 的定义域是 ()。

The domain of the function $y = \frac{\sqrt{x+3}}{x-2} + 3\log x$ is () .

- A. $[-3, +\infty)$
- B. $(-3, 0) \cup (0, 2) \cup (2, +\infty)$
- C. $(0, 2) \cup (2, +\infty)$
- D. $[-3, 0) \cup (0, 2) \cup (2, +\infty)$

答案 Answer: C

解析 Analysis:

中文：定义域需同时满足：根号内 $x+3 \geq 0$ ，分母 $x-2 \neq 0$ ，对数真数 $x > 0$ ，取三者交集得 $x > 0$ 且 $x \neq 2$ 。

英文：The domain requires $x+3 \geq 0$ under the square root, $x-2 \neq 0$ for the denominator, and $x > 0$ for the logarithm. The intersection of the three conditions gives $x > 0$ and $x \neq 2$.

第 36 题 Question 36

若 $\alpha \in [\frac{3\pi}{2}, 2\pi]$ ，且 $\cos \alpha = \frac{3}{5}$ ，则 $\sin \frac{\alpha}{2} = ()$ 。

If $\alpha \in [\frac{3\pi}{2}, 2\pi]$ and $\cos \alpha = \frac{3}{5}$, then $\sin \frac{\alpha}{2} = ()$.

- A. $\frac{\sqrt{5}}{5}$
- B. $\pm \frac{\sqrt{5}}{5}$
- C. $\pm \frac{2\sqrt{5}}{5}$
- D. $\frac{2\sqrt{5}}{5}$

答案 Answer: A

解析 Analysis:

中文：由 α 的范围得 $\frac{\alpha}{2} \in [\frac{3\pi}{4}, \pi]$ ，因此 $\sin \frac{\alpha}{2} > 0$ ；由半角公式 $\sin \frac{\alpha}{2} = \sqrt{\frac{1 - \cos \alpha}{2}}$ ，代入计算得 $\frac{\sqrt{5}}{5}$ 。

英文：From the range of α , we have $\frac{\alpha}{2} \in [\frac{3\pi}{4}, \pi]$, so $\sin \frac{\alpha}{2} > 0$. By the half-angle

formula $\sin \frac{\alpha}{2} = \sqrt{\frac{1-\cos \alpha}{2}}$, substitution gives $\frac{\sqrt{5}}{5}$.

第 37 题 Question 37

关于诱导公式，下列表达式正确的是 ()。

Regarding the trigonometric reduction formulas, which of the following expressions is correct?

- A. $\sin (\pi + \alpha) = \sin \alpha$
- B. $\cos (2\pi - \alpha) = \cos \alpha$
- C. $\tan (\pi + \alpha) = -\tan \alpha$
- D. $\sin (\frac{\pi}{2} + \alpha) = \sin \alpha$

答案 Answer: B

解析 Analysis:

中文：余弦函数是偶函数且周期为 2π ，因此 $\cos (2\pi - \alpha) = \cos (-\alpha) = \cos \alpha$ ，该式正确；其余选项均存在符号错误。

英文：Cosine is an even function with period 2π , so $\cos (2\pi - \alpha) = \cos (-\alpha) = \cos \alpha$. This expression is correct, while the others have sign errors.

第 38 题 Question 38

已知 $\cos \alpha = -\frac{1}{2}$, $\alpha \in (\frac{\pi}{2}, \pi)$, 则 $\sin \frac{\alpha}{2}$ 的值是 ()。

Given $\cos \alpha = -\frac{1}{2}$ and $\alpha \in (\frac{\pi}{2}, \pi)$, the value of $\sin \frac{\alpha}{2}$ is ().

- A. $-\frac{1}{2}$
- B. $-\frac{\sqrt{3}}{2}$
- C. $\frac{1}{2}$
- D. $\frac{\sqrt{3}}{2}$

答案 Answer: D

解析 Analysis:

中文：由 α 的范围得 $\frac{\alpha}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$ ，因此 $\sin \frac{\alpha}{2} > 0$ ；由半角公式 $\sin \frac{\alpha}{2} = \sqrt{\frac{1-\cos \alpha}{2}}$ ，代入计算得 $\frac{\sqrt{3}}{2}$ 。

英文：From the range of α , we have $\frac{\alpha}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$, so $\sin \frac{\alpha}{2} > 0$. By the half-angle formula

$\sin \frac{\alpha}{2} = \sqrt{\frac{1-\cos \alpha}{2}}$, substitution gives $\frac{\sqrt{3}}{2}$.

第 39 题 Question 39

下列关于函数 $y = \tan(-2x)$ 的说法，正确的是 ()。

Which of the following statements about the function $y = \tan(-2x)$ is correct?

- A. 该函数在定义域上有界 (The function is bounded over its domain)
- B. 该函数定义域是 $\mathbb{R}$ (The domain of the function is $\mathbb{R}$)
- C. 该函数的最小正周期为 π (The minimum positive period of the function is π)
- D. 该函数在 $(-\frac{\pi}{4}, \frac{\pi}{4})$ 上是单调递减的 (The function is monotonically decreasing on $(-\frac{\pi}{4}, \frac{\pi}{4})$)

答案 Answer: D

解析 Analysis:

中文: $y = \tan(-2x) = -\tan 2x$; $x = \pm \frac{\pi}{4}$ 不在定义域内, 在开区间 $(-\frac{\pi}{4}, \frac{\pi}{4})$ 内, $\tan 2x$ 单调递增, 因此 $-\tan 2x$ 单调递减。

英文: $y = \tan(-2x) = -\tan 2x$. $x = \pm \frac{\pi}{4}$ are not in the domain. On the open interval $(-\frac{\pi}{4}, \frac{\pi}{4})$, $\tan 2x$ is monotonically increasing, so $-\tan 2x$ is monotonically decreasing.

第 40 题 Question 40

若椭圆 C 的中心在原点, 焦点在 y 轴上, 焦距为 4, 离心率为 $\frac{2}{3}$, 则 C 的方程是 ()。

If the center of ellipse C is at the origin, foci are on the y -axis, the focal length is 4, and the eccentricity is $\frac{2}{3}$, the equation of C is ().

- A. $\frac{x^2}{12} + \frac{y^2}{8} = 1$
- B. $\frac{x^2}{9} + \frac{y^2}{5} = 1$
- C. $\frac{x^2}{5} + \frac{y^2}{9} = 1$
- D. $\frac{x^2}{8} + \frac{y^2}{12} = 1$

答案 Answer: C

解析 Analysis:

中文: 焦距 $2c = 4$ 得 $c = 2$, 由离心率 $e = \frac{c}{a} = \frac{2}{3}$ 得 $a = 3$, $b^2 = a^2 - c^2 = 5$; 焦点在 y 轴, 方程为 $\frac{x^2}{5} + \frac{y^2}{9} = 1$ 。

英文: Focal length $2c = 4$ gives $c = 2$. Eccentricity $e = \frac{c}{a} = \frac{2}{3}$ gives $a = 3$, and $b^2 = a^2 - c^2 = 5$. With foci on the y -axis, the equation is $\frac{x^2}{5} + \frac{y^2}{9} = 1$.

第 41 题 Question 41

若数列 $\{a_n\}$ 的前 n 项和 $S_n = n^3 + n^2 + n$, 则通项 $a_n = ()$ 。

If the sum of the first n terms of the sequence $\{a_n\}$ is $S_n = n^3 + n^2 + n$, then the general term $a_n = ()$.

- A. $3n^2 - n + 1$

- B. $n^2 + n + 1$
 C. $3n^2 + n + 1$
 D. $3n^2 + 2n + 1$

答案 Answer: A

解析 Analysis:

中文：当 $n \geq 2$ 时， $a_n = S_n - S_{n-1}$ ，代入化简得 $a_n = 3n^2 - n + 1$ ；验证 $n = 1$ 时也满足该式，因此为此数列的通项公式。

英文：For $n \geq 2$, $a_n = S_n - S_{n-1}$. Simplification gives $a_n = 3n^2 - n + 1$, which also holds for $n = 1$. So this is the general term of the sequence.

第 42 题 Question 42

关于对数函数下面说法正确的是 ()。

Which of the following statements about logarithmic functions is correct?

- A. $\log_2 3 < \log_3 5$
 B. 若 $\log_a x > \log_a x^2 (a > 1)$, 则 $0 < x < 1$ (If $\log_a x > \log_a x^2 (a > 1)$, then $0 < x < 1$)
 C. $\log_{100} 10 = \log_{10} 1$
 D. 若 $\log_a x > \log_b x$, 则 $a > b$ (If $\log_a x > \log_b x$, then $a > b$)

答案 Answer: B

解析 Analysis:

中文：当 $a > 1$ 时，对数函数单调递增，因此不等式等价于 $x > x^2$ ，结合真数大于 0 解得 $0 < x < 1$ ，该选项正确。英文：When $a > 1$, the logarithmic function is monotonically increasing, so the inequality is equivalent to $x > x^2$. Combined with the condition that the argument is positive, we get $0 < x < 1$. The option is correct.

1, the logarithmic function is monotonically increasing, so the inequality is equivalent to $x > x^2$. Combined with the condition that the argument is positive, we get $0 < x < 1$. The option is correct.

第 43 题 Question 43

下列关于椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a, b > 0)$ 的说法，正确的是 ()。

Which of the following statements about the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a, b > 0)$ is correct?

- A. 该椭圆焦点在 x 轴 (The foci of the ellipse are on the x -axis)
 B. 该椭圆上任意一点到椭圆两个焦点的距离之和等于 $2a$ (The sum of the distances from any point on the ellipse to the two foci is equal to $2a$)
 C. 该椭圆的中心在原点 (The center of the ellipse is at the origin)
 D. 该椭圆的离心率 $e = \frac{b}{a}$ (The eccentricity of the ellipse is $e = \frac{b}{a}$)

答案 Answer: C

解析 Analysis:

中文：该方程为椭圆的标准形式，中心在坐标原点；题干仅标注 $a, b > 0$ ，未约定 a 为

长半轴长，因此无法确定焦点所在轴与长轴长度；椭圆离心率为 $e = \frac{c}{a}$ (a 为长半轴长)，因此只有 C 选项正确。

英文：This is the standard equation of an ellipse with center at the origin. The problem only states $a, b > 0$ and does not specify that a is the semi-major axis, so we cannot determine the position of foci or the length of the major axis. The eccentricity of an ellipse is $e = \frac{c}{a}$ (where a is the semi-major axis). Thus only option C is correct.

第 44 题 Question 44

已知平面向量 $\vec{a}$, $\vec{b}$, $\vec{c}$ 满足 $|\vec{a}| = 1$, $|\vec{b}| = 1$, $\vec{a} \cdot \vec{b} = -\frac{1}{2}$,

且 $\vec{c} = \lambda\vec{a} + (1 - \lambda)\vec{b} (\lambda \in \mathbb{R})$, 则 $|\vec{c}|$ 的最小值为 ()。

Given plane vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ satisfy $|\vec{a}| = 1$, $|\vec{b}| = 1$, $\vec{a} \cdot \vec{b} = -\frac{1}{2}$, and $\vec{c} = \lambda\vec{a} + (1 - \lambda)\vec{b} (\lambda \in \mathbb{R})$, the minimum value of $|\vec{c}|$ is () .

A. $\frac{1}{2}$

B. $\frac{\sqrt{3}}{2}$

C. 1

D. $\frac{\sqrt{2}}{2}$

答案 Answer: A

解析 Analysis:

中文：将 $|\vec{c}|^2$ 展开为关于 λ 的二次函数 $3\lambda^2 - 3\lambda + 1$ ，其最小值为 $\frac{1}{4}$ ，因此 $|\vec{c}|$ 的最小值为 $\frac{1}{2}$ 。

英文：Expand $|\vec{c}|^2$ to get the quadratic function $3\lambda^2 - 3\lambda + 1$ in λ . Its minimum value is $\frac{1}{4}$, so the minimum of $|\vec{c}|$ is $\frac{1}{2}$.

第 45 题 Question 45

设等比数列 $\{a_n\}$ 的前 n 项和为 S_n ，且 $2S_n = 3a_{n+1} - 3 (n = 1, 2, \dots)$ ，则 $S_n = ()$ 。

Let the sum of the first n terms of the geometric sequence $\{a_n\}$ be S_n , and $2S_n = 3a_{n+1} - 3 (n = 1, 2, \dots)$, then $S_n = ()$.

A. $\frac{2}{3}(\frac{3}{5})^n - \frac{2}{3}$

B. $\frac{3}{2}(\frac{5}{3})^n - \frac{3}{2}$

C. $\frac{3}{2}(\frac{3}{5})^n - \frac{3}{2}$

D. $\frac{3}{2}(\frac{5}{3})^n - \frac{2}{3}$

答案 Answer: B

解析 Analysis:

中文：由递推式求得首项 $a_1 = 1$ ，公比 $q = \frac{5}{3}$ ，代入等比数列求和公式得 $S_n = \frac{3}{2}(\frac{5}{3})^n - \frac{3}{2}$ 。

英文：From the recurrence relation, we get the first term $a_1 = 1$ and common ratio $q = \frac{5}{3}$. Substitute into the sum formula of geometric sequence: $S_n = \frac{3}{2}(\frac{5}{3})^n - \frac{3}{2}$.

第 46 题 Question 46

若复数 $z = -1 + i$ ，其中 i 是虚数单位，则 $\frac{z+1}{z+2} + \frac{z+2}{z+1} = ()$ 。

If the complex number $z = -1 + i$, where i is the imaginary unit, then $\frac{z+1}{z+2} + \frac{z+2}{z+1} = ()$.

A. $\frac{3}{2} + \frac{1}{2}i$

B. 2

C. $\frac{3}{2} - \frac{1}{2}i$

D. 1

答案 Answer: C

解析 Analysis:

中文：代入 $z = -1 + i$ 得 $z + 1 = i$ ， $z + 2 = 1 + i$ ，分别化简两个分式后相加，计算结果为 $\frac{3}{2} - \frac{1}{2}i$ 。

英文：Substitute $z = -1 + i$ to get $z + 1 = i$ and $z + 2 = 1 + i$. Simplify each fraction and add them up, the result is $\frac{3}{2} - \frac{1}{2}i$.

第 47 题 Question 47

已知 $\{a_n\}$ 满足 $a_1 = \frac{1}{2}$ ， $a_{n+1} = \frac{a_n}{1+2a_n}$ ，令 $b_n = \frac{1}{a_n}$ ，则 $S_n = b_1 + b_2 + \cdots + b_n = ()$ 。

Given $\{a_n\}$ satisfies $a_1 = \frac{1}{2}$ and $a_{n+1} = \frac{a_n}{1+2a_n}$, let $b_n = \frac{1}{a_n}$, then $S_n = b_1 + b_2 + \cdots + b_n = ()$.

A. $n(n+1)$

B. $n(n-1)$

C. $\frac{n^2}{2}$

D. n^2

答案 Answer: A

解析 Analysis:

中文：对递推式两边取倒数得 $b_{n+1} = b_n + 2$ ，即 $\{b_n\}$ 是首项为 2、公差为 2 的等差数列，前 n 项和为 $n(n+1)$ 。

英文：Take reciprocals on both sides of the recurrence to get $b_{n+1} = b_n + 2$. So $\{b_n\}$ is an arithmetic sequence with first term 2 and common difference 2, and its sum of first n terms is $n(n+1)$.

第 48 题 Question 48

已知有 n 个编号互不相同的白球和 n 个编号互不相同的红球。现从中随机取出 n 个球，设取出球的颜色交替出现的概率为 $P(n)$ ，这里的颜色交替是指取出球的 n 个球的颜色为：第 1 个球为白色，第 2 个球为红色，第 3 个球为白色，第 4 个球为红色，以此类推；或第 1 个球为红色，第 2 个球为白色，第 3 个球为红色，第 4 个球为白色，以此类推。则下列关系正确的是 ()。

Given n white balls with distinct numbers and n red balls with distinct numbers. Now n balls are randomly drawn, and let $P(n)$ be the probability that the colors of the drawn balls alternate. Here, color alternation means the colors of the n drawn balls are: the 1st ball is white, the 2nd is red, the 3rd is white, the 4th is red, and so on; or the 1st ball is red, the 2nd is white, the 3rd is red, the 4th is white, and so on. Which of the following relations is correct?

- A. $P(2) > P(4) > P(6)$
- B. $P(2) > P(6) > P(4)$
- C. $P(6) > P(4) > P(2)$
- D. $P(4) > P(2) > P(6)$

答案 Answer: A

解析 Analysis:

中文：按取出顺序计算交替排列的情况，分别计算得 $P(2) = \frac{2}{3}$, $P(4) = \frac{6}{35}$,

$P(6) = \frac{10}{231}$ ，概率随 n 增大单调递减，因此 $P(2) > P(4) > P(6)$ 。

英文：Calculate the alternating arrangements in the order of drawing. The results are

$P(2) = \frac{2}{3}$, $P(4) = \frac{6}{35}$, $P(6) = \frac{10}{231}$. The probability decreases monotonically as n increases, so $P(2) > P(4) > P(6)$.